Problem 2. Consider a cone with base radius r and height h. What is the average area $A(x)$ of the vertical cross section at x of the cone, as x ranges over $[-r, r]$?

Solution. Letting $A(x)$ be the area of the vertical cross section at x, the average is computed as follows:

$$\frac{1}{2r}\int_{-r}^{r} A(x)\mathrm{d}x$$

The integral

$$\int_{-r}^{r} A(x)\mathrm{d}x$$

is the volume of the cone, as it would be computed by vertical cross sections. Thus, it comes out to $\pi r^2 h/3$, and the average comes out to $\pi r h/6$, after dividing by $2r$.